

Stellar Ages, Abundances and Classification: Optimizing Machine-Learning Models to Accurately Predict Stellar Parameters

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ABSTRACT

The age of stars and stellar populations provide useful information with respect to the evolution of galaxies, including the Milky Way. Unfortunately, the estimation of this parameters comes with several limitations regarding instrumentation and techniques that are transversal to different populations of stars. This work uses the GALAH DR4 survey (Buder et al. 2024) and machine-learning tools to optimize model parameters so that it can accurately predict: membership of stars (whether they belong to either disk of the Milky Way or the halo), abundances of [Mg/Fe], and stellar ages. The sample is cleaned and reduced to approximately 430,000 stars. The metrics for every model are generally favorable, resulting in accuracies of 0.73, 0.78 and 0.86 for the thin disk, thick disk, and halo respectively. The [Mg/Fe] model returned a regression score of $R^2 = 0.89$ with a mean-square-error (MSE) of 0.002. Finally, the chemical clock performs well with a score of $R^2 = 0.96$ and $MSE = 0.38$. For the first two models, the most important parameters agree with relationships found in literature. The chemical clock, however, uses Li abundances after finding no correlation with [Y/Fe] and age, which hinders a proper explanation. Linear relationships between abundances and age are required to improve the chemical clock.

Keywords: Milky Way - Thick Disk - Thin Disk - Halo - Machine Learning - Chemical Clock - Metallicity

1. INTRODUCTION

Stellar ages are an important parameter that grant insight into the history of galaxies (Shejeelammal et al. 2024). There are methods, such as isochrone fitting, along with other techniques like asteroseismology, that allow for the estimation of this parameter. In particular, stellar populations let us uncover new information about the evolution and formation of the Milky Way, for example Hayden et al. (2022). There are several limitations that come with these methods, such as the ability to perform these estimates on a certain type of stars, such as Main-Sequence stars, or even technological limits. The latter problem comes from a accuracy and precision in instruments that is reserved for certain sections of the sky Hayden et al. (2022) thus barring off an entire set of stars whose age cannot be guessed.

Newer techniques involve the analysis of linear relationships between age and abundance of stars that are close by, which is referred to as a chemical clock. In particular, it has been discovered that the yttrium abun-

dance, [Y/Fe], provides a linear correlation with the age (Da Silva et al. 2012), and later, Vázquez et al. (2022) discovered that the functionality of this relationship depends on the galactocentric radius R_{GC} . Other relationships have been explored, such as the relationship between [Mg/Fe] and [Fe/H], as well as $[\alpha/\text{Fe}]$, that have proven to be abundances that can be used to separate the different structures of the Milky Way (thick disk, thin disk, and halo). α corresponds to alpha-elements. This paper will use a similar definition of $[\alpha/\text{Fe}]$ as Hayden et al. (2022), which states that it is an average of the abundance of Si, Mg, Ti I and can be represented as such:

$$[\alpha/\text{Fe}] = \log \left(\frac{n(\alpha)}{n(\text{Fe})} \right)_* - \log \left(\frac{n(\alpha)}{n(\text{Fe})} \right)_\odot \quad (1)$$

Where $n(\alpha)$ corresponds to:

$$n(\alpha) = n(\text{Si}) + n(\text{Ti}) + n(\text{Mg}) \quad (2)$$

There have been efforts to use machine-learning to estimate the age of stars, like in [Moya et al. \(2022\)](#), to construct these chemical clocks. There are routines available, notably by `scipy` [Virtanen et al. \(2020\)](#), that allow for prediction of parameter values by training a classifier or regression model with real datasets.

The goal of the following work is to determine the optimal parameters to produce to a machine-learning model that produces the best predictions of MW structure membership, abundances of $[\text{Mg}/\text{Fe}]$, and stellar age, as well as discovering the stellar parameters that are most important in each model. The paper is divided into the following sections: section 2, which will detail the preparation of the dataset, separation of the dataset, and training methodology of each section. Section 3, which contains the results of the MW structure membership model, and $[\text{Mg}/\text{Fe}]$ prediction model, and the chemical clock. Finally, section 4 will provide a review of the results of this report, how they were determined, and suggestions to improve this study in the future.

2. METHODS

2.1. Data

The data was taken from the fourth GALAH survey release which, based on a decade of observations, provides the chemical abundances of up to 32 elements for 917,588 stars along with several other properties from the GAIA satellite [Buder et al. \(2024\)](#). Aside from the abundances, this paper makes use of the survey’s dynamical and orbital parameters. Originally, the sample contains up to 10^6 stars.

2.2. Dataset Preparation

The criteria used to prepare the dataset were the following: firstly, the selection of parameters that will be used throughout the work. This includes stellar parameters such as t_{eff} , $\log g$, $v \sin i$, mass, age, and bolometric luminosity. Most chemical abundances were considered, excluding those who had mostly unavailable data ($> 60\%$ rows filled with NaN values) such is the case of the rapid neutron capture elements (Ru, Sm, Eu) and some light and heavy slow neutron capture elements (Mo, Ba, La, Ce, Nd), along with their respective flags. The tables were also filtered by those stars which had a signal-to-noise per pixel for the spectrum from each camera > 30 , as recommended by the GALAH documentation. The tables containing the stellar parameters, abundances, and dynamical and orbital parameters were merged into one table.

Outliers were removed from the sample by applying sigma-clipping since the distribution of most param-

eters corresponded to a gaussian distribution. The final dataset resulted in approximately 430,000 stars.

2.2.1. Separation by Structure

Before beginning the training algorithm, the sample was separated into three categories or classes: halo (H) thin disk, (D) and thick disk (TD). This separation was done by using a Toomre diagram ([Toomre 1964](#)) and considering the method provided in [da Silva et al. \(2023\)](#). This method consists of using the total velocity V_{tot} of each star, which is given by:

$$V_{\text{tot}} = (U_{\text{LSR}}^2 + V_{\text{LSR}}^2 + W_{\text{LSR}}^2)^{1/2} \quad (3)$$

Where LSR corresponds to Local Standard Rest. The limits to determine the membership of each star to each galactic structure are the following: stars with $V_{\text{tot}} < 50 \text{ km s}^{-1}$ are considered D stars. Stars with $50 \text{ km s}^{-1} < V_{\text{tot}} < 180 \text{ km s}^{-1}$ are TD stars. Finally, H stars are those with $V_{\text{tot}} > 180 \text{ km s}^{-1}$. After making this separation, flags of ones and zeros were created to indicate whether or not a star belonged to a certain structure of the Milky Way (MW) and added to the main dataset as columns.

2.2.2. Training Overview

The machine-learning aspect was done using the `sci-kit` ([Pedregosa et al. 2011](#)) package and its modules `DecisionTreeClassifier`, `RandomForestClassifier`, and `RandomForestRegressor`. A random forest is an estimator that fits a number of decision tree classifiers on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control overfitting, according to the documentation. The algorithm allows several parameters to handle the creation of the tree; among them, is there depth of the tree, number of estimators which indicate the number of trees, and the minimum number of samples required to split an internal node. The first of these parameters will be studied closely. To decide on a value of maximum tree depth, an analysis of depth versus precision, accuracy, and recall was performed. The definition of these metrics are the following, where TP represents true positives, TN represents true negatives, FP , false positives, and FN , false negatives:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

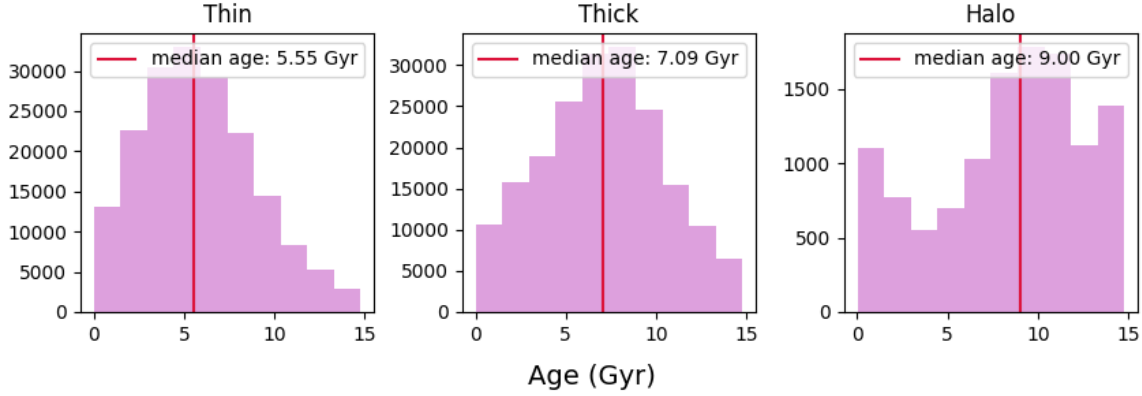


Figure 1. The final age characteristics of each structure in the MW. It is in agreement with the characteristics of these parts found in literature.

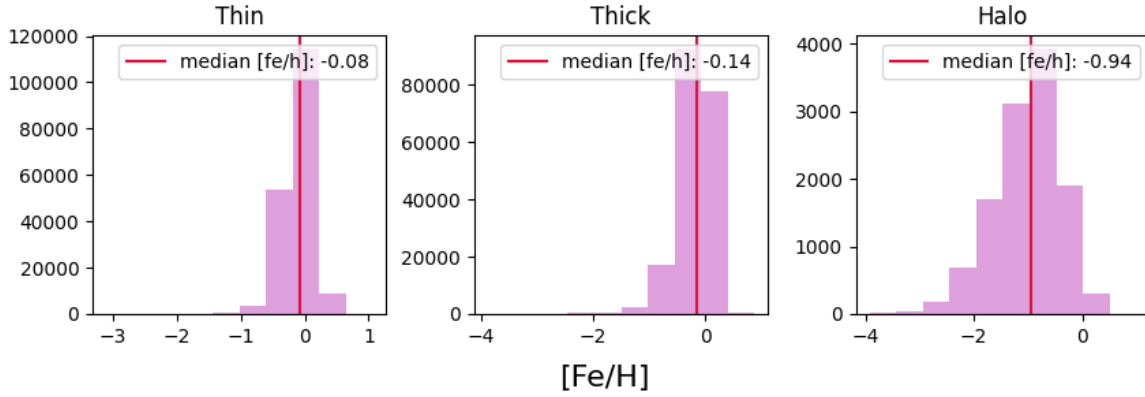


Figure 2. The final metallicity characteristics of each structure in the MW. It is in agreement with the characteristics of these parts found in literature.

This work focuses on maximizing precision given that there are no major consequences regarding having elevated false negatives. In this case, the priority is the correctness of the model. The precision, accuracy, and recall percentages are interpreted through a confusion matrix to ensure that the diagonal values that represent TN and TP are elevated, which indicates high precision. Several other features of the training model were altered to achieve better results.

Beforehand, the training model splits the whole data into a test sample, which will be used to carry out the actual training, and the data itself that the model will use based on results of the training. Varying the percentage of the data which will be used for testing has an effect over the precision of the model.

Prior to supplying the model with the data, a balancing or levelling out of the data was performed. This was necessary given that, the number stars in the halo are several orders of magnitudes less than the stars in the thin and thick disks. For this reason, the rows containing stars belonging in the halo were duplicated until

achieving a number that was relatively similar to the amount of stars that belong in the disks.

After running the data through the classifier models, the importance of every feature was evaluated to determine the three that had the most impact.

2.2.3. Training $[Mg/Fe]$ Model

A similar process was done for creating a model that predicts the values of $[Mg/Fe]$, with regards to balancing the data and eliminating outliers. This time, however, a regression model was used instead of a classifier. The model was trained by using chemical abundances that did not involve alpha elements, such as Si and Ti. The metrics involved are the regression score R^2 , which indicates the correctness of the model, and the closer to one, the better. This value, alongside the mean-square-error (MSE), were used to evaluate the optimal depth of the tree.

2.2.4. Training Chemical Clock

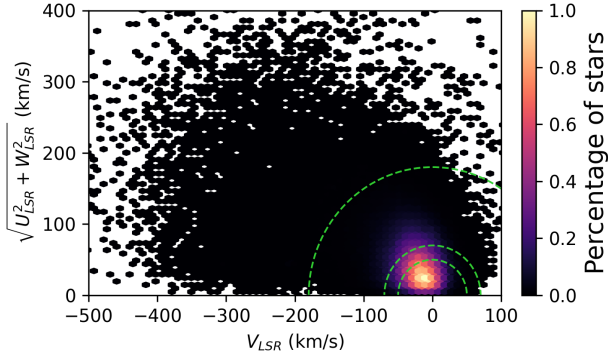


Figure 3. Toomre diagram of the dataset. The largest circumference represents the limit between halo (outer region) and thick disk stars (inner region), and the smaller circumference designates all those that are within the thin disk (inner region). The gap between the smallest circumference and the second largest corresponds to an overlap region of thin and thick disk stars.

Like the $[Mg/Fe]$ model, a regression was used to predict the ages of the stars. This consisted in firstly observing the abundances that correlated most with the age, and using these in the model.

3. RESULTS & DISCUSSION

3.1. *D, TD, and H Membership*

By using the Toomre diagram in Fig. 3, a separation was performed, resulting in approximately 191,548 TD stars, 181,911 D stars, and 11877 halo stars. For this reason, the balancing of halos was carried out. The metrics for the D, TD, and H stars can be found in Tables 1, and the confusion matrix for the D stars can be found in 4. The rest of the confusion matrixes for the other structures can be found in the appendix in figures 10 and 11.

	Precision	Accuracy	Recall
D	0.55	0.73	0.84
TD	0.66	0.78	0.66
H	0.81	0.86	0.80

Table 1. Model metrics for the predictions of each structure.

The three most important stellar parameters or abundances that were determined for the can be found in Tables 2, 3, and 4 respectively.

According to Table 1 and the information given in the confusion matrix, it's clear that the model performs well in general, given that the accuracy of every every model achieves well above 50%. The precision and recall being high also indicates the the models are able to minimize

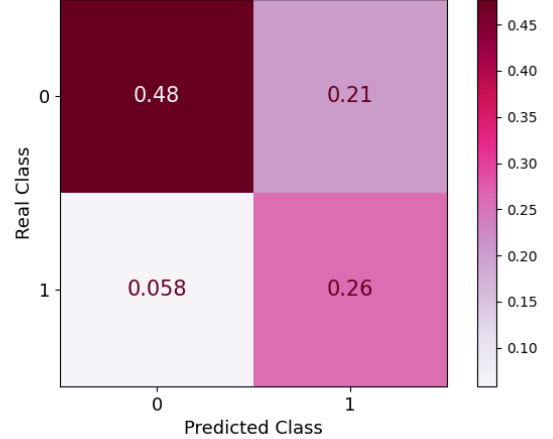


Figure 4. Confusion matrix for D stars.

Parameter	Importance
$[Si/Fe]$	0.18
$[Mg/Fe]$	0.14
$\log g$	0.12

Table 2. The three important parameters found for the **thin disk** along with their value of importance.

Parameter	Importance
$[Si/Fe]$	0.09
$\log g$	0.09
$[Fe/H]$	0.08

Table 3. The three important parameters found for the **thick disk** along with their value of importance.

Parameter	Importance
$[Si/Fe]$	0.19
$[Fe/H]$	0.18
$[Ca/Fe]$	0.11

Table 4. The three important parameters found for the **halo** along with their value of importance.

false positives as well as correctly detect true positives. This is mostly due to the correct selection of parameters for each of the trees, and in particular, the maximum depth, as can be seen in Fig. 5, which plays a great difference in the precision.

Finally, note that each structure share similar abundances that are most important, such as $[Si/Fe]$, $[Fe/H]$ and $[Mg/Fe]$. This is in agreement with the findings in Adibekyan et al. (2012), which state that there exists a separation between the thin and thick

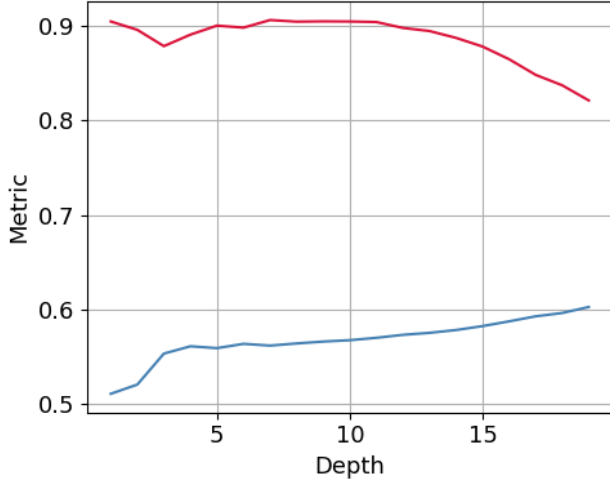


Figure 5. Precision (red) and recall (blue) versus tree depth for the thin disk model. Equal plots were created for all three of the models.

disks due to the chemical abundance of alpha elements (which include, Si, Mg, and Ti). Therefore, that one of the alpha-element abundances rank so high as an important parameter is to be expected.

3.2. Prediction of $[Mg/Fe]$

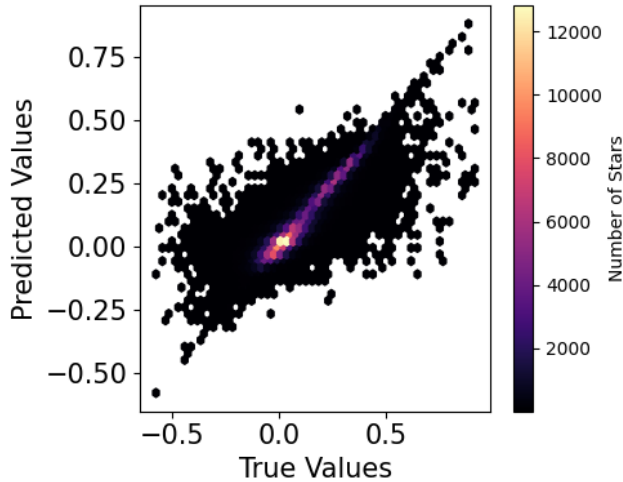


Figure 6. The predicted values of the $[Mg/Fe]$ chemical abundances versus the true values.

The model for the prediction of $[Mg/Fe]$ finished with a regression score of $R^2 = 0.89$, with $MSE = 0.002$, which indicates excellent performance of the model. This score was achieved by analyzing the depth versus metric score which can be seen in Fig. 7. This result can also be reflected by the linear relation between predicted values and true values in Fig. 6.

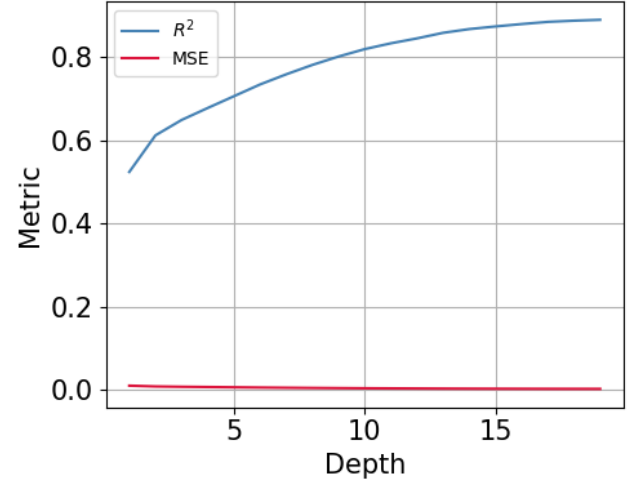


Figure 7. The values of R^2 and MSE with the variation of the tree depth. Note the increase of the regression score with the increase of depth.

The three most important parameters that were could in this part of the report can be found in Table 5. Shejeelammal et al. (2024), point out the direct relation between $[Mg/Fe]$ and $[Fe/H]$, the latter happening to be the most important parameter of the three. This concurs with the two populations of $[Mg/Fe]$ at low-metallicity, the high- α or thick disk and the low- α or thin disk that are pointing out in Fig. 13 and Fig. 14 in the appendix where a bifurcation is distinguishable. Shejeelammal et al. (2024) states that this is due to a difference in chemical and dynamical evolutionary trajectories as well as the star formation history of the MW disk.

Parameter	Importance
$[Fe/H]$	0.63
$[Si/Fe]$	0.04
$[Ca/Fe]$	0.03

Table 5. The three most important parameters for the prediction of $[Mg/Fe]$, listed from most to least important.

3.3. Chemical Clock

The model for the chemical clock resulted in a regression score $R^2 = 0.96$ and a $MSE = 0.38$ which indicates that the model is predicting excellently. This is also reflected in the linear relation between the predicted values and the true values in Fig. 8. In order to select the chemical abundances that would be used for the clock, the relations between age and abundances were analyzed using Fig. 12. In this case, most abundances seem to give little detail regarding the age of stars save for Li

Parameter	Importance
[Li/Fe]	0.20
[Ti/Fe]	0.15
[Sc/Fe]	0.07

Table 6. The three most important parameters for the chemical clock in decreasing order of importance.

and Sc. Instead of only using these two, however, the author considered all elements to avoid heavy biases by only using two chemical abundances. What’s to be expected is a negative correlation with elements such as Y and O, the former being used in works such as [Shejeelammal et al. \(2024\)](#). In this case, it’s Li that offers most information about the age of the stars.

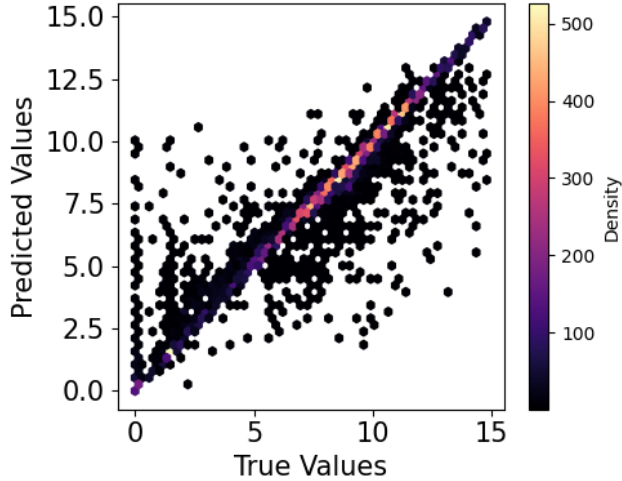


Figure 8. The predicted values from the model versus the true values. The linear relation further proves that the model’s predicting is accurate.

It’s important to note that for this section there was a secondary filtering of the data according to those established by [Hayden et al. \(2022\)](#), who did a similar experiment, which involved using $T_{eff} < 6200K$. This is because, like the authors encountered, there are issues for abundances measured for hotter stars.

The parameters for the model, particularly maximum tree depth, was chose by using the relationship between depth and R^2 which can be seen in Fig. 9. For this part, a tree depth of 19 was selected.

The three most important parameters found in this section can be found in Table 6. It’s clear the [Li/Fe] provided the most information regarding the age of the stars.

There is little literature with regards to Li abundances playing part in the age of stars, as the abundance of Li

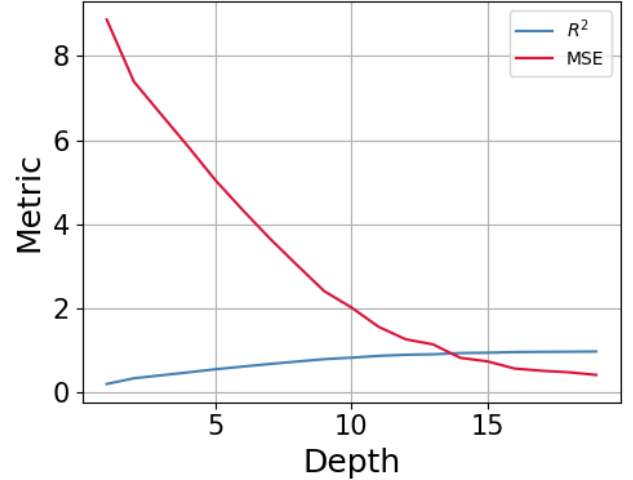


Figure 9. Relation between R^2 and MSE , and the maximum tree depth. It’s clear the score improves as the depth of the tree increases.

itself still provides a problem in modern-day astronomy in the halo ([Fields 2011](#)). Regardless, the model was able to function well by using these three parameters to estimate the age of stars.

4. CONCLUSION

The present work sought to train classification and regression models with the optimal parameters in order to: firstly, correctly separate a set of stars into a thin disk, thick disk, or halo category. Secondly, to create a model that could accurately predict the [Mg/Fe] abundance in stars. And thirdly, to predict the ages of stars with $T_{eff} < 6200K$. A proper analysis of the model parameters was performed to optimize the accuracy, specifically by plotting the tree depth against different metrics. Given the results shown throughout the paper, the goal was achieved with accuracy scores of 0.73, 0.79 and 0.86 for the thin disk, thick disk, and halo respectively, along with regression scores of $R^2 = 0.89$ and $R^2 = 0.96$ for the [Mg/Fe] prediction and chemical clock respectively. Important features for the MW structure separation included [Si/Fe], [Mg/Fe] and [Ti/Fe], which is in agreement with findings by [Adibekyan et al. \(2012\)](#). The important features for the [Mg/Fe] estimation, which mainly relies on [Fe/H], is supported by evidence in [Shejeelammal et al. \(2024\)](#) that indicates the connection between these two abundances. The report was not without it’s problems after having found that a lack of relation between Y and the age of the stars hindered a proper explanation for the usage of the Li abundance to predict this parameter. To improve the final parts of this work, it would be useful to have a

chemical abundance that displayed a direct correlation with the age, be it positive or negative. Some of the metrics for the models could also be improved, for example, the precision of the thin disk model. This could happen by further altering the parameters of the classifier model until the metrics reach higher values.

Software: astropy ([Astropy Collaboration et al. 2013, 2018](#)), Cloudy ([Ferland et al. 2013](#)), Source Extractor ([Bertin & Arnouts 1996](#))

APPENDIX

A. JUPYTER NOTEBOOK

A Jupyter Notebook with the calculations can be found as an attached file alongside this report.

B. CONFUSION MATRIXES

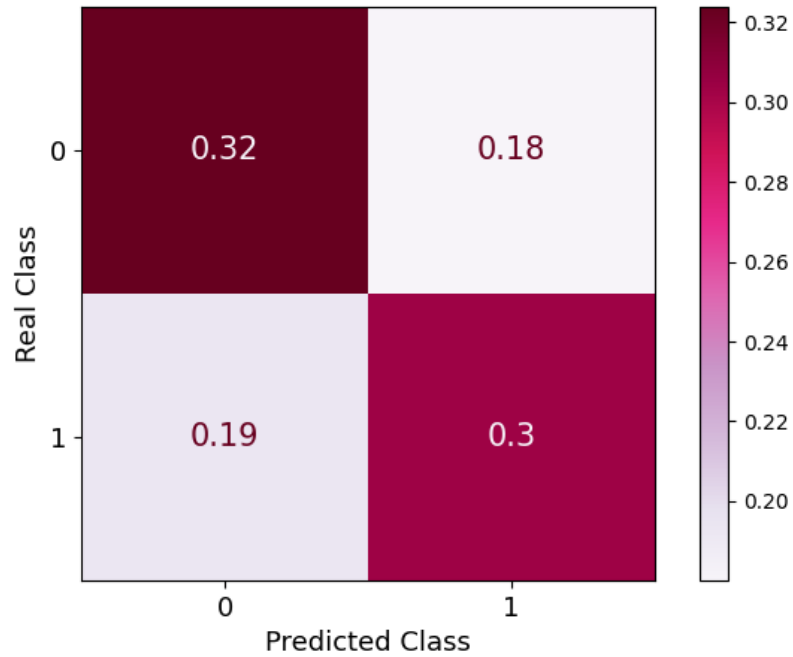


Figure 10. Confusion matrix for TD stars.

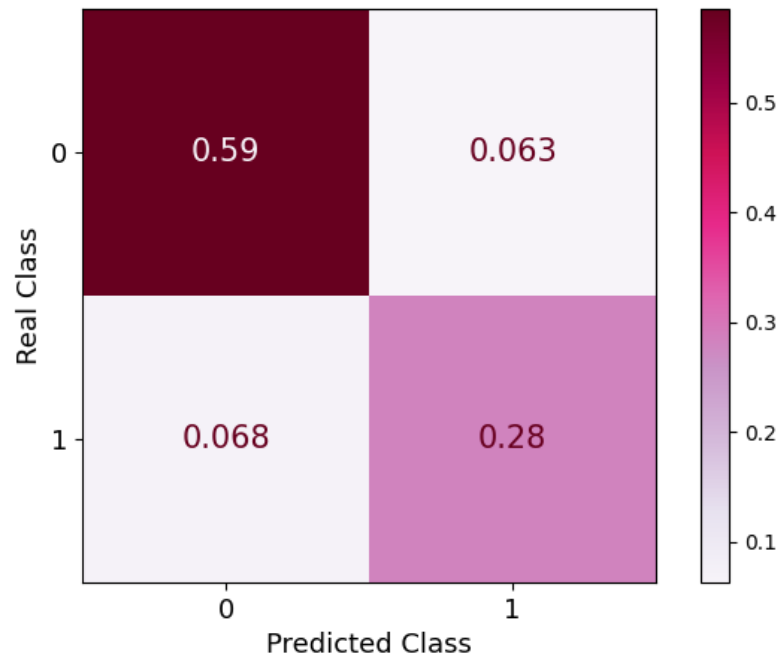


Figure 11. Confusion matrix for H stars.

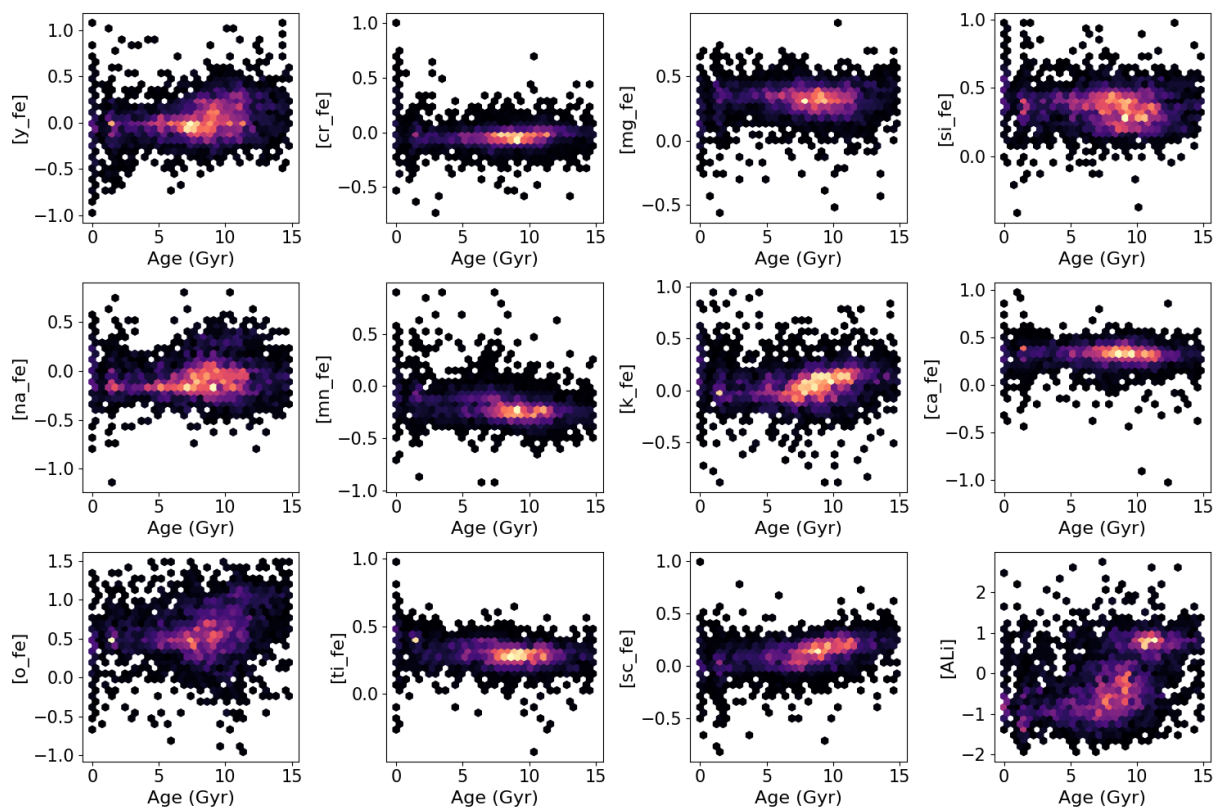


Figure 12. Correlation between age and the chemical abundances involved in the dataset.

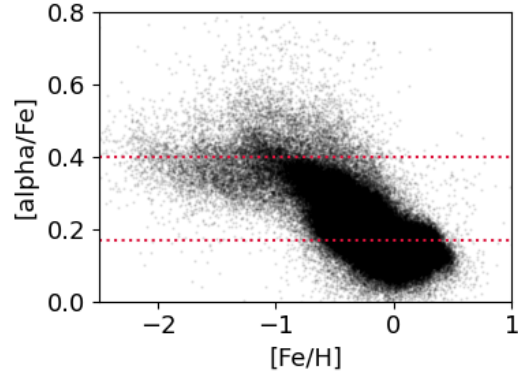


Figure 13. Relationship between $[\text{Fe}/\text{H}]$ and alpha elements. A bifurcation in the red dotted lines can be appreciated, where the superior dotted lines indicated a separation between the thick and thin disk. Due to the large amount of data, this separation is not so clear.

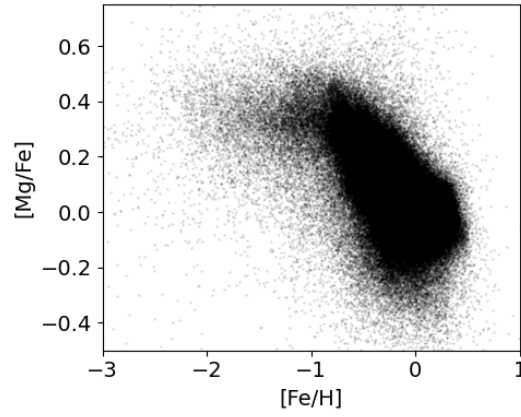


Figure 14. Relationship between $[\text{Fe}/\text{H}]$ and $[\text{Mg}/\text{Fe}]$. Like with the alpha elements, a bifurcation can be discerned, indicating a separation between the thick and thin disk. Due to the large amount of data, this separation is not so clear.

C. CHEMICAL ABUNDANCE VS. AGE

D. SEPARATION OF THE DISKS BY CHEMICAL ABUNDANCE

REFERENCES

- Adibekyan, V. Z., Sousa, S., Santos, N., et al. 2012, *Astronomy & Astrophysics*, 545, A32
- Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, *A&A*, 558, A33, doi: [10.1051/0004-6361/201322068](https://doi.org/10.1051/0004-6361/201322068)
- Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, *AJ*, 156, 123, doi: [10.3847/1538-3881/aabc4f](https://doi.org/10.3847/1538-3881/aabc4f)
- Bertin, E., & Arnouts, S. 1996, *A&AS*, 117, 393, doi: [10.1051/aas:1996164](https://doi.org/10.1051/aas:1996164)
- Buder, S., Kos, J., Wang, E. X., et al. 2024, The GALAH Survey: Data Release 4. <https://arxiv.org/abs/2409.19858>
- da Silva, M., de Lima, M., Velloso, E., & do Nascimento, J.-D. 2023, *The Astrophysical Journal*, 958, 32
- Da Silva, R., de Mello, G. P., Milone, A., et al. 2012, *Astronomy & Astrophysics*, 542, A84
- Ferland, G. J., Porter, R. L., van Hoof, P. A. M., et al. 2013, *RMxAA*, 49, 137. <https://arxiv.org/abs/1302.4485>
- Fields, B. D. 2011, *Annual Review of Nuclear and Particle Science*, 61, 47
- Hayden, M. R., Sharma, S., Bland-Hawthorn, J., et al. 2022, *Monthly Notices of the Royal Astronomical Society*, 517, 5325
- Moya, A., Sarro, L. M., Delgado-Mena, E., et al. 2022, *Astronomy & Astrophysics*, 660, A15
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. 2011, *Journal of Machine Learning Research*, 12, 2825. <http://jmlr.org/papers/v12/pedregosa11a.html>
- Shejeelammal, J., Meléndez, J., Rathsam, A., & Martos, G. 2024, *Astronomy & Astrophysics*, 690, A107
- Toomre, A. 1964, *Astrophysical Journal*, vol. 139, p. 1217-1238 (1964)., 139, 1217
- Vázquez, C. V., Magrini, L., Casali, G., et al. 2022, *Astronomy & Astrophysics*, 660, A135
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, *Nature methods*, 17, 261